

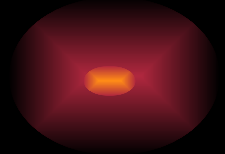


DETERMINATION OF SAMPLE SIZE

SAMPLE SIZE AND ITS DETERMINATION



- In sampling analysis the most ticklish question is: What should be the size of the sample or how large or small should be ' n '?



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- ***If the sample size ('n') is too small, it may not serve to achieve the objectives;***

and if it is too large, we may incur huge cost and waste resources.

- **As a general rule, one can say that**
- **the sample must be of an optimum size i.e., it should neither be excessively large nor too small.**

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- **Technically, the sample size should be large enough to give a confidence interval of desired width**
 - **And**
 - **as such the size of the sample must be chosen by some logical process before sample is taken from the universe.**

- **Size of the sample should be determined by a researcher keeping in view the following points:**





- **(i) *Nature of universe:* Universe may be either homogenous or heterogenous in nature. Technically, this can be termed as the dispersion factor.**
- **(ii) *Number of classes proposed:* *Small sized sample* might not be able to give a reasonable number of items in each class-group.**

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- (iii) ***Nature of study: Intensive/extensive*** ***If items are to be intensively and continuously studied, the sample should be small.***
 - (iv) ***Type of sampling: Sampling technique*** ***plays an important part in determining the size of the sample.***
 - **A small random sample is apt to be much superior to a larger but badly selected sample.**

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- ***(v) Standard of accuracy and acceptable confidence level:***
 - **The More the accuracy desired, the larger the sample shall be.**
 - **To double the accuracy for a fixed significance level, the sample size has to be increased fourfold.**

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- **(vi) *Availability of finance:***
 - **(vii) *Other considerations:***
 - ***Nature of the units,***
 - ***size of the population,***
 - ***size of questionnaire,***
 - ***Availability of trained investigators,***

- **The time available for completion of the study are a few other considerations**



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- **There are two alternate approaches for determining the size of the sample.**
 - **The first approach**
 - **is “to specify the precision of estimation desired and then to determine the sample size necessary to insure it”**

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- **the second approach “uses Bayesian statistics to weigh the cost of additional information against the expected value of the additional information.”**

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- **The first approach is capable of giving a mathematical solution, and hence is frequently used**
 - ***The limitation-***
 - **It does not analyse the cost of gathering information *vis-a-vis* the *expected* value of information**

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- **The second approach is theoretically optimal, but it is seldom used because of**
 - **the difficulty involved in measuring the value of information.**
 - **Hence, we shall mainly concentrate**
 - **here on the first approach.**

THE FIRST APPROACH

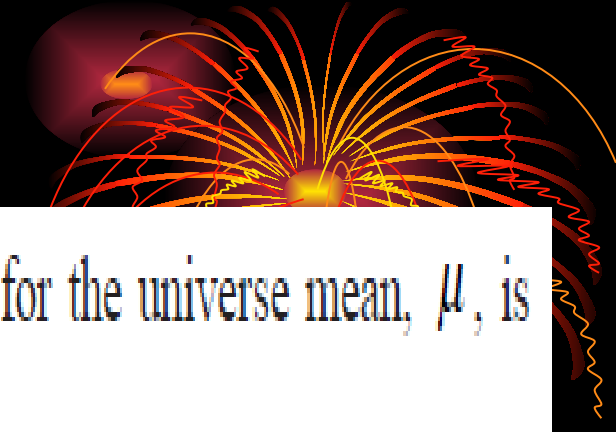
**DETERMINATION OF SAMPLE SIZE
THROUGH THE APPROACH
BASED ON PRECISION RATE AND
CONFIDENCE LEVEL**



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- **Researcher will have to specify the precision**
 - **that he wants in respect of his estimates concerning the population parameters.**
 - **For instance, a researcher may like to estimate the mean of the universe within ± 3 of the true mean**
 - **with 95 per cent confidence.**



- **In this case we will say that the desired precision is ± 3 , i.e.,**
- **if the sample mean is 100, the true value of the mean will be no less than 97 and no more than 103. In other words, all this means that the acceptable error, e , is equal to 3.**



(a) *Sample size when estimating a mean:* The confidence interval for the universe mean, μ , is given by

$$\bar{X} \pm z \frac{\sigma_p}{\sqrt{n}}$$

where $\bar{X}$ = sample mean;

z = the value of the standard variate at a given confidence level (to be read from the table giving the areas under normal curve as shown in appendix) and it is 1.96 for a 95% confidence level;

n = size of the sample;

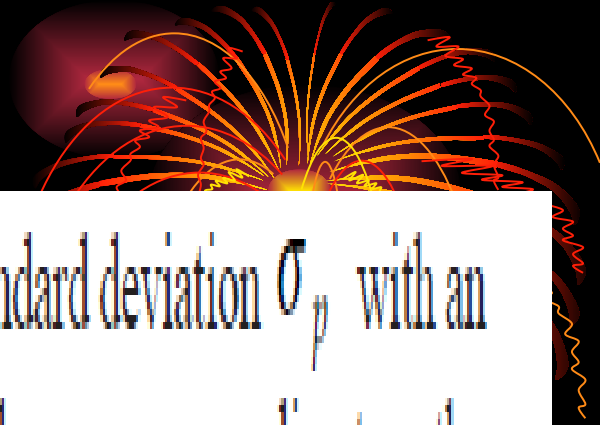


σ_p = standard deviation of the population (to be estimated from past experience or on the basis of a trial sample). Suppose, we have $\sigma_p = 48$ for our purpose.

If the difference between μ and $\bar{x}$ or the acceptable error is to be kept within ± 3 of the sample mean with 95% confidence, then we can express the acceptable error, 'e' as equal to

$$e = z \cdot \frac{\sigma_p}{\sqrt{n}} \text{ or } 3 = 1.96 \frac{48}{\sqrt{n}}$$

$$\text{Hence, } n = \frac{(1.96)^2 (48)^2}{(3)^2} = 9.834 \approx 10.$$



In a general way, if we want to estimate μ in a population with standard deviation σ_p with an error no greater than ' e ' by calculating a confidence interval with confidence corresponding to z , the necessary sample size, n , equals as under:

$$n = \frac{z^2 \sigma^2}{e^2}$$

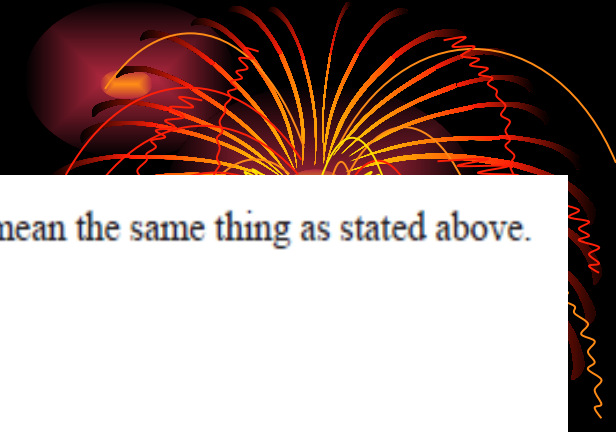


All this is applicable whe the population happens to be infinite. Bu in case of finite population, the above stated formula for determining sample size will become

$$n = \frac{z^2 \cdot N \cdot \sigma_p^{2*}}{(N-1)e^2 + z^2 \sigma_p^2}$$

* In case of finite population the confidence interval for μ is given by

$$\bar{X} \pm z \frac{\sigma_p}{\sqrt{n}} \times \sqrt{\frac{(N-n)}{(N-1)}}$$



where $\sqrt{(N-n)/(N-1)}$ is the finite population multiplier and all other terms mean the same thing as stated above.

If the precision is taken as equal to 'e' then we have

$$e = z \frac{\sigma_p}{\sqrt{n}} \times \sqrt{\frac{(N-n)}{(N-1)}}$$

or

$$e^2 = z^2 \frac{\sigma_p^2}{n} \times \frac{N-n}{N-1}$$

or

$$e^2 (N-1) = \frac{z^2 \sigma_p^2 N}{n} - \frac{z^2 \sigma_p^2 n}{n}$$

or

$$e^2 (N-1) + z^2 \sigma_p^2 = \frac{z^2 \sigma_p^2 N}{n}$$

or

$$n = \frac{z^2 \cdot \sigma_p^2 \cdot N}{e^2 (N-1) + z^2 \sigma_p^2}$$

or

$$n = \frac{z^2 \cdot N \cdot \sigma_p^2}{(N-1)e^2 + z^2 \sigma_p^2}$$

This is how we obtain the above stated formula for determining 'n' in the case of infinite population given the precision and confidence level.

$$\hat{\mu} = \frac{z^2 \cdot N \cdot \sigma_p^2}{(N-1)e^2 + z^2 \sigma_p^2}$$





where

N = size of population

n = size of sample

e = acceptable error (the precision)

σ_p = standard deviation of population

z = standard variate at a given confidence level.

Illustration

- **Determine the size of the sample for estimating the true weight of the cereal containers for the universe with $N = 5000$ on the basis of the following information:**



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- **(1) the variance of weight = 4 ounces on the basis of past records.**
 - **(2) estimate should be within 0.8 ounces of the true average weight with 99% probability.**
 - **Will there be a change in the size of the sample if we assume infinite population in the given case? If so, explain by how much?**

Solution:

- ***In the given problem we have the following:***
- ***$N = 5000$;***
- ***$S_p = 2$ ounces (since the variance of weight = 4 ounces);***
- ***$e = 0.8$ ounces (since the estimate should be within 0.8 ounces of the true average weight);***



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- ***$z = 2.57$ (as per the table of area under normal curve for the given confidence level of 99%).***

- **Hence, the confidence interval for μ is given by**



$$\bar{X} \pm z \cdot \frac{\sigma_p}{\sqrt{n}} \cdot \sqrt{\frac{N-n}{N-1}}$$



and accordingly the sample size can be worked out as under:

$$\begin{aligned} n &= \frac{z^2 \cdot N \cdot \sigma_p^2}{(N - 1) e^2 + z^2 \sigma_p^2} \\ &= \frac{(2.57)^2 \cdot (5000) \cdot (2)^2}{(5000 - 1) (0.8)^2 + (2.57)^2 (2)^2} \\ &= \frac{132098}{3199.36 + 26.4196} = \frac{132098}{3225.7796} = 40.95 \cong 41 \end{aligned}$$



Hence, the sample size (or n) = 41 for the given precision and confidence level in the above question with finite population. But if we take population to be infinite, the sample size will be worked out as under:


$$n = \frac{z^2 \sigma_p^2}{e^2}$$

$$= \frac{(2.57)^2 (2)^2}{(0.8)^2} = \frac{26.4196}{0.64} = 41.28 \simeq 41$$

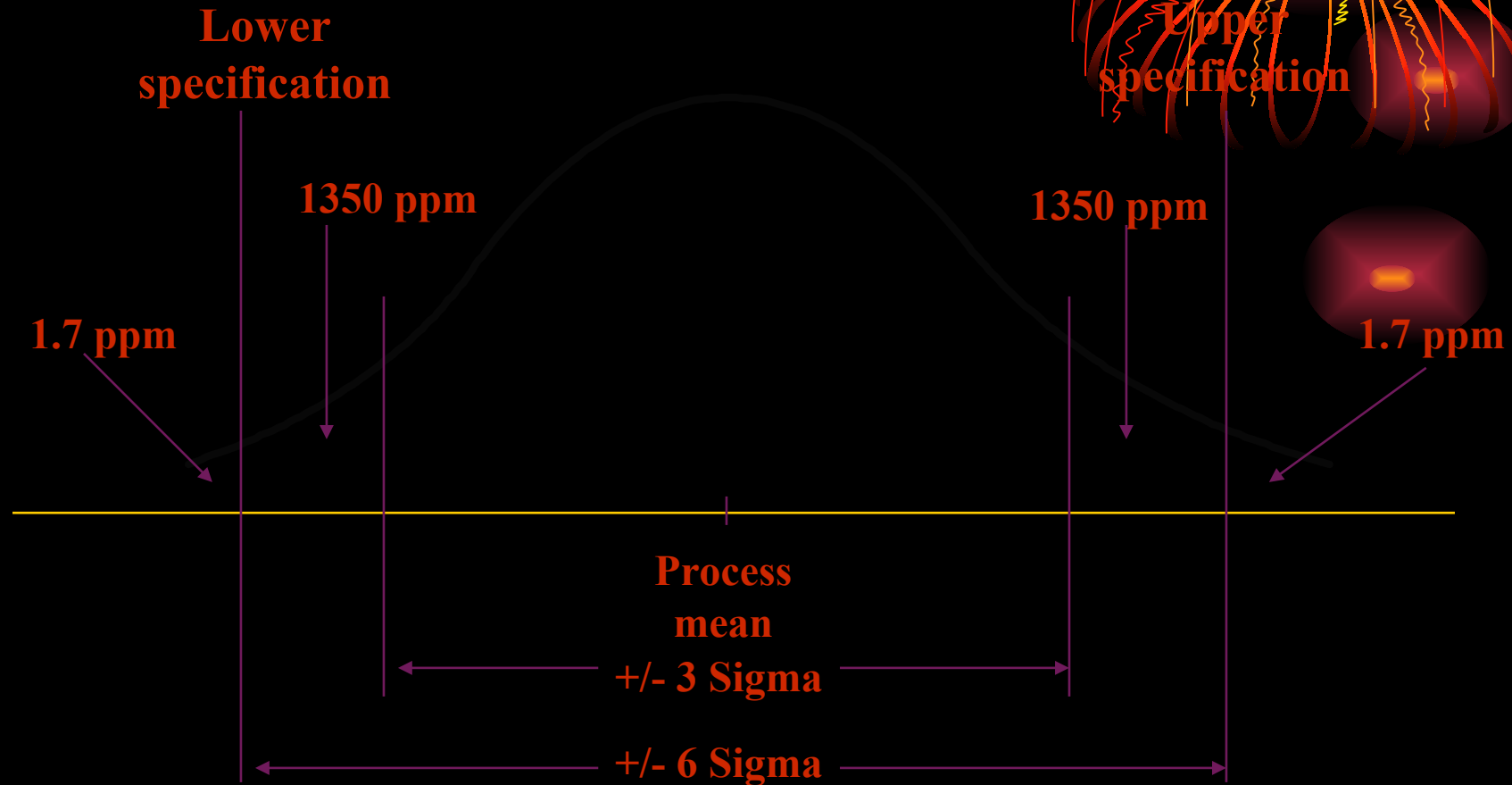
PRESUMPTION OF STANDARD DEVIATION OF POPULATION

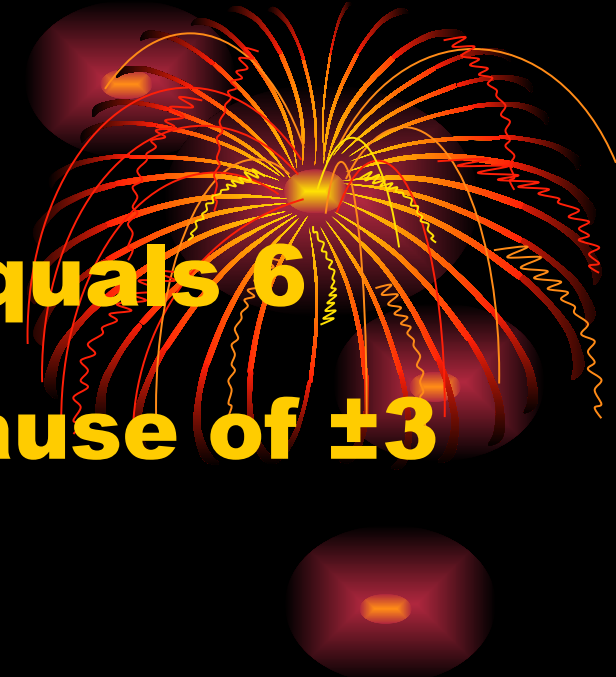


- **Generally the standard deviation of the population is not pre-known.**
 - **Then, if, the range of values are known, a crude estimate of the standard deviation of the population can be used to get the said estimate of standard deviation as follows:**
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- **Since 99.7 per cent of the area under normal curve lies within the range of ± 3 standard deviations,**
 - **we may make the assumptions as follow:**
 - **These limits include almost all of the distribution.**

3 Sigma and 6 Sigma Quality



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- **Hence the given range equals 6 standard deviations because of ± 3 SD.**

- **Thus, a rough estimate of the population standard deviation would be:**



$6\hat{\sigma} = \text{the given range}$

$$\hat{\sigma} = \frac{\text{the given range}}{6}$$



If the range happens to be, say Rs 12, then

$$\hat{\sigma} = \frac{12}{6} = \text{Rs } 2.$$

and this estimate of standard deviation, $\hat{\sigma}$, can be used to determine the sample size in the formulae stated above.



THANK YOU